Pratheek Nathani

Parkland, FL | 954-247-8348 | nathanipratheek@gmail.com | LinkedIn: pratheek-nathani

EDUCATION

UNIVERSITY OF FLORIDA Gainesville, FL
B.S. Computer Science, B.S. Mathematics Expected May 2028
Cumulative GPA: 4.0/4.0; 7x AIME Qualifier (221 USAMO Index)
Relevant Coursework: Operating Systems, Data Structures & Algorithms, Discrete Math, Linear Algebra, Sets & Logic

WORK EXPERIENCE

Microsoft Redmond, WA
Software Engineer Intern May 2026 – August 2026

- Extending hardware root-of-trust verification to next-generation confidential computing silicon, implementing attestation flows and firmware validation for pre-release hardware (details under NDA).
- Built an AI agent with an MCP server automating root-cause analysis of Azure Confidential VM SLA misses — mapping errors to root causes and automating cross-team escalation — cutting on-call triage time 45% across 12 engineers.
- Trained the agent's underlying classification models on failure telemetry to attribute host failures to specific firmware build regressions at 98% recall, replacing multi-day manual log investigation.

Modernizing Medicine Boca Raton, FL
Software Engineer Intern May 2025 – May 2026

- Developed an agentic AI scheduling system using Python, LangChain, and FastAPI to autonomously process 1M+ daily patient messages via EHR APIs, reducing scheduling overhead across a 70M-user platform.
- Architected containerized FastAPI microservices with Docker sustaining 700 req/min, implementing OAuth2, async request processing, and feature flag-controlled ML rollouts for zero-downtime deployments.
- Built 10+ automated test suites via CircleCI CI/CD pipelines achieving 92% coverage, validating agent behavior across edge case patient scheduling scenarios and ML deployment conditions to prevent erroneous responses from reaching production.
- Deployed a patient-facing voice agent using ElevenLabs and Twilio, architecting an MCP layer with 15 tools wrapping 50+ endpoints and a knowledge base to handle patient scheduling and clinical queries, reducing manual call volume by 21%.

AlgoGators Investment Fund Gainesville, FL
Senior Quant Developer August 2024 – Present

- Built quantitative risk models implementing Conditional VaR, Fama-French factor exposures, and Information Ratio calculations in Python, feeding directly into the execution system to drive live trading decisions on a $500k portfolio.
- Developed a futures position sizing optimization model adjusting allocations based on real-time market risk conditions across futures and equities strategies, reducing average slippage by ~10-15% in both live trading and backtests.
- Refactored critical trading systems from Python to C++ achieving 3x+ improvements in real-time market data processing and database latency, validated against live portfolio and backtest benchmarks.

Endgrate Cambridge, MA
Software Engineer Intern May 2024 – August 2024

- Engineered 500+ real-time API integrations in Python, building standardized connectors across CRM, ERP, and helpdesk platforms to enable unified data access through a single API, reducing customer integration time by ~20%.
- Designed and deployed API endpoints with asynchronous data fetching, error handling, and normalization pipelines, ensuring reliable data synchronization across 100+ third-party services and customer applications.

PROJECTS

Chaos Engineering Platform | Go, Python, Kubernetes, Prometheus

- Built a chaos engineering platform in Go and Python injecting fault types (pod kill, network partition, CPU stress) into AWS/Kubernetes clusters, with a real-time dashboard tracking error rates and latency across 12 failure scenarios.
- Implemented a blast radius governor that enforces per-experiment resource limits and triggers automated rollbacks when SLO thresholds are breached, reducing mean recovery time by 60% versus manual intervention during simulated outages.

ADDITIONAL

Languages: Python, C++, Java, C, JavaScript, TypeScript, Go, Rust, HTML, CSS, SQL
Tools: LangChain, LangGraph, FastAPI, Docker, MLflow, CircleCI, Jenkins, Git, Feature Flags, ElevenLabs, Twilio, Sidekiq
APIs/Libraries: OpenAI, Pandas, NumPy, SciPy, Pydantic, Pytorch, RAG, Healthcare APIs (FHIR)
Leadership/Activities: Susquehanna Sophomore Discovery Day (In-Person), Susquehanna Freshman Discovery Day (Virtual)